VHP: Verkle-Verified Hypergraph Provenance for Trustworthy AI Decision Systems

Cryptographic End-to-End Verifiability for
Agentic AI Reasoning over Domain Knowledge

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Abstract

AI systems deployed in safety-critical domains such as healthcare and finance must provide auditable, verifiable decision traces. Current agentic AI systems — including retrieval-augmented generation (RAG) pipelines and tool-using reasoning agents — cannot cryptographically prove what knowledge was used, whether that knowledge was untampered, or how iterative reasoning steps causally influenced the final decision. We propose **VHP (Verkle-verified Hypergraph Provenance)**, a three-layer verification architecture that provides end-to-end cryptographic guarantees for AI decision systems: (1) a *Hypergraph Knowledge Layer* that represents multi-way domain interactions (e.g., drug-drug-condition polypharmacy risks) that pairwise knowledge graphs cannot capture; (2) a *Verkle Verification Layer* that provides constant-size (∼96 byte) integrity proofs over hypergraph substructures, achieving ∼94% proof size reduction compared to Merkle trees; and (3) a *Provenance DAG Layer* that records the full causal chain of an agentic reasoning loop, cryptographically linking every reasoning step to verified knowledge and prior inferences. VHP is reasoning-engine agnostic — it wraps any AI system (LLM, SLM, or hybrid) with verifiability guarantees without modifying the model itself. We evaluate on a healthcare drug interaction task using DrugBank (1.4M+ interactions) and demonstrate that VHP adds less than 4% computational overhead while enabling complete, independently verifiable audit trails. To our knowledge, this is the first work combining Verkle trees, hypergraph knowledge representation, and reasoning provenance DAGs for trustworthy AI.

Keywords: Verkle Trees, Hypergraph, Provenance, Trustworthy AI, Agentic Reasoning, Knowledge Verification, Healthcare AI

1 Introduction

The deployment of AI reasoning systems in regulated domains imposes requirements that go far beyond accuracy. When an AI-powered clinical decision support system recommends against prescribing a medication, regulatory frameworks (HIPAA, EU AI Act [European Parliament, 2024], FDA AI/ML guidelines [FDA, 2021]) demand answers to three fundamental questions:

1. **What knowledge?** — What was the state of the knowledge base when the decision was made, and can we prove it was untampered?
2. **What combinations?** — Were multi-factor risks (drug + drug + condition + patient demographics) evaluated, not just pairwise interactions?
3. **What reasoning?** — What was the exact chain of reasoning that led to the conclusion, and can each step be independently verified?

No existing AI architecture answers all three questions with cryptographic guarantees. Current systems have fundamental gaps:

Retrieval-Augmented Generation (RAG) systems [Lewis et al., 2020] retrieve text chunks and feed them to a language model, but provide no proof that the retrieved chunks are authentic or that the knowledge base was not modified between retrieval and response.

Graph-based RAG systems [Edge et al., 2024, He et al., 2024] improve context quality through graph structure but use pairwise knowledge graphs that cannot represent multi-way interactions — the combination {Warfarin, Aspirin, CKD Stage 3, Age > 65} carries polypharmacy risk that no single pairwise edge captures.

Agentic reasoning systems (ReAct [Yao et al., 2023], tool-using agents) perform iterative think-act-observe loops but produce flat text logs with no cryptographic linkage between steps and no proof that earlier observations influenced later reasoning.

We address these gaps with **VHP (Verkle-verified Hypergraph Provenance)**, a three-layer verification architecture:

1. **Hypergraph Knowledge Layer** represents domain knowledge using hyperedges that connect arbitrary sets of entities, capturing multi-way interactions that pairwise graphs miss (Section 3.2).
2. **Verkle Verification Layer** provides constant-size cryptographic integrity proofs over hypergraph substructures using vector commitments, reducing proof sizes by $\sim$94% compared to Merkle trees (Section 3.3).
3. **Provenance DAG Layer** records the full causal graph of an agentic reasoning loop, where each node (thought, observation, conclusion) is cryptographically linked to the verified knowledge it consumed and the prior reasoning steps it depended on (Section 3.4).

Critically, VHP is **reasoning-engine agnostic**. It does not modify or constrain the AI model — it wraps the model's inputs and outputs with verification metadata. Whether the reasoning engine is GPT-4o, Phi-3, Qwen-2.5, or a future model, the verification stack is identical.

1.1 Contributions

1. **Hypergraph Knowledge Representation for AI:** We formalize multi-way domain interactions as hyperedges and show their advantage over pairwise knowledge graphs for polypharmacy risk detection (Section 3.2).

2. **Verkle-Verified Knowledge Integrity:** We adapt Verkle trees [Kuszmaul, 2019] — previously studied only in blockchain contexts — to provide constant-size integrity proofs for AI knowledge bases (Section 3.3).

3. **Provenance DAG for Agentic Reasoning:** We define a cryptographically linked directed acyclic graph that captures the causal structure of iterative AI reasoning, enabling independent audit verification (Section 3.4).

4. **Unified Audit Protocol:** We specify a complete audit record format that combines all three layers, enabling end-to-end verification of AI decisions (Section 4).

5. **Prototype and Evaluation:** We implement VHP and evaluate on healthcare drug interaction detection using DrugBank data (Section 5).

2 Related Work

2.1 Knowledge Graphs and Hypergraphs

Knowledge graphs represent domain knowledge as entity-relation-entity triples [Pan et al., 2024]. While effective for pairwise relationships, they cannot natively represent multi-way interactions.

Hypergraphs [Bretto, 2013] generalize graphs by allowing edges to connect arbitrary sets of vertices. Recent work has explored hypergraphs for drug interaction modeling [Feng et al., 2020] and recommendation systems [Yu et al., 2021], but no prior work integrates hypergraphs with cryptographic verification for AI systems.

2.2 Verified Data Structures

Merkle trees [Merkle, 1979] provide integrity verification through hierarchical hashing but suffer from proof sizes that grow with tree width. Verkle trees [Kuszmaul, 2019] replace hash-based siblings with polynomial commitments, achieving constant-size proofs. Ethereum is adopting Verkle trees for stateless client validation [Buterin, 2021]. Our work is the first to apply Verkle trees to AI knowledge base verification.

2.3 AI Provenance and Explainability

Existing approaches to AI transparency focus on feature attribution (SHAP [Lundberg and Lee, 2017], LIME [Ribeiro et al., 2016]) or attention visualization. These explain *why* a model made a decision but cannot cryptographically prove *what knowledge* the decision was based on. Data provenance systems in databases [Buneman et al., 2001] track data lineage but do not address iterative AI reasoning chains. Our Provenance DAG extends data provenance concepts to agentic AI reasoning with cryptographic guarantees.

2.4 Agentic AI Systems

ReAct [Yao et al., 2023] introduced the think-act-observe loop for language model reasoning. Tool-using agents [Schick et al., 2024] extend this with external tool calls. Multi-agent systems [Wu et al., 2023, Hong et al., 2023] decompose tasks across specialized agents. AgentRouter [AgentRouter, 2025] uses GNNs for routing queries to agents. None of these systems provide cryptographic verification of the knowledge used or the reasoning trace. VHP can wrap any of these architectures with verifiability.

3 VHP Architecture

3.1 Overview

VHP consists of three layers, each addressing a distinct verification requirement:

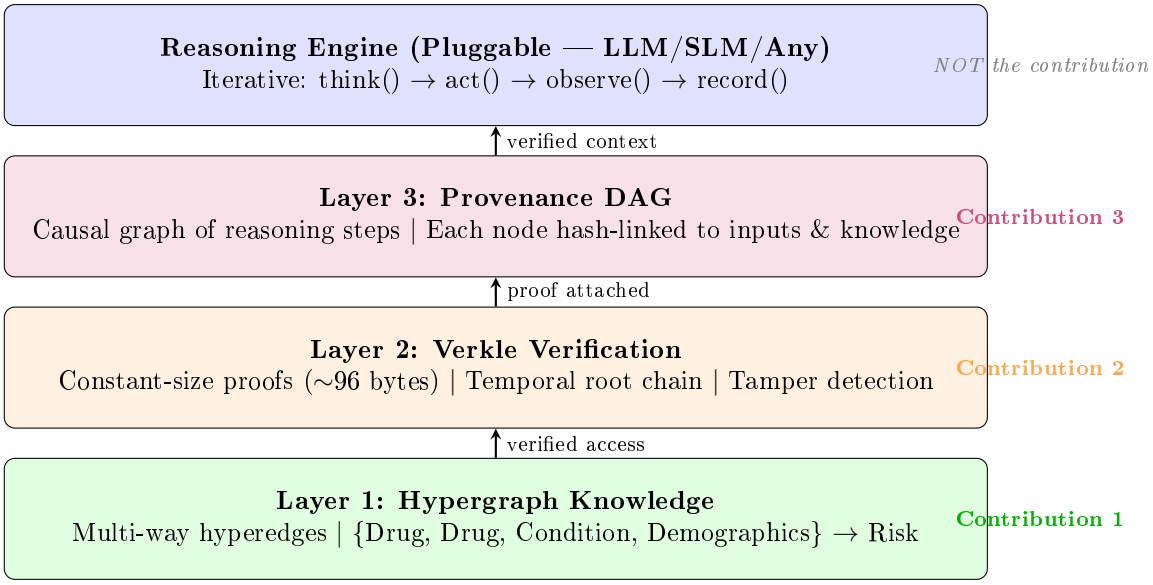

Figure 1: VHP three-layer verification architecture. The reasoning engine is pluggable and not part of the contribution. The three verification layers (Hypergraph, Verkle, Provenance DAG) are the novel contributions.

3.2 Layer 1: Hypergraph Knowledge Representation

3.2.1 Motivation

Traditional knowledge graphs represent facts as triples (s, r, o) where s is a subject entity, r is a relation, and o is an object entity. This pairwise structure cannot represent multi-way interactions:

- *Drug polypharmacy:* The combination {Warfarin, Aspirin, CKD Stage 3} creates a bleeding risk that no pairwise edge between any two of these entities captures.
- *Demographic-adjusted risk:* {Metformin, CKD, eGFR < 30, Age > 65} is contraindicated, but {Metformin, CKD, eGFR > 45, Age < 50} may be acceptable with dose adjustment.

3.2.2 Formal Definition

A *domain hypergraph* is a tuple $\mathcal{H} = (V, E_2, E_h, \mathcal{T}, \phi)$ where:

- V is a set of entities (drugs, conditions, demographics)
- $E_2 \subseteq V \times \mathcal{R} \times V$ is a set of pairwise edges (standard KG triples)
- $E_h \subseteq 2^V \times \mathcal{L}$ is a set of hyperedges, where each $e_h = (S, \ell)$ connects a subset $S \subseteq V$ with $|S| \geq 2$ and carries a label $\ell \in \mathcal{L}$ (e.g., risk level, severity)
- $\mathcal{T} : V \to \text{Types}$ is an entity type function
- $\phi : E_h \to \mathbb{R}$ is a confidence/severity scoring function

This representation subsumes standard knowledge graphs (E_2) while adding multi-way interactions (E_h).

3.2.3 Canonical Serialization

For cryptographic verification, hyperedges must have a deterministic byte representation:

$$\text{serialize}(e_h) = \text{sort}(\{v_i\}_{v_i \in S}) \| \ell \| \phi(e_h) \tag{1}$$

Sorting entity IDs ensures that $\{A, B, C\}$ and $\{C, A, B\}$ produce identical hashes.

3.3 Layer 2: Verkle Verification

3.3.1 Motivation

To verify knowledge integrity, we need compact proofs that a specific piece of knowledge was part of the verified knowledge base. Merkle trees require $O((d-1)\log_d n)$ sibling hashes per proof, where d is the branching factor. For a knowledge base with $n = 10{,}000$ hyperedges and $d = 256$, this yields $\sim$255 hashes per proof level.

Verkle trees [Kuszmaul, 2019] replace hash-based verification with polynomial (vector) commitments, where proof size is independent of the branching factor:

$$\text{Proof}_{\text{Merkle}}(d,n) = (d-1)\log_d n \text{ hashes} \quad \text{vs} \quad \text{Proof}_{\text{Verkle}}(d,n) = \log_d n \text{ commitments}+1 \text{ opening} \tag{2}$$

3.3.2 Construction

We partition the hypergraph into k semantic subgroups $\{G_1, \ldots, G_k\}$ (e.g., drug-drug interactions, drug-condition contraindications, treatment protocols). Each subgroup is serialized canonically and committed:

$$C_i = \text{Commit}(\text{serialize}(G_i)) \quad \forall i \in [k] \tag{3}$$

These commitments form the leaves of a Verkle tree. The root commitment C_{root} serves as the cryptographic fingerprint of the entire knowledge base.

3.3.3 Proof Generation and Verification

To prove that subgroup G_i is part of the verified knowledge base:

$$\pi_i = \text{VerkleProof}(C_i, C_{\text{root}}) \quad |\pi_i| = O(1) \approx 96 \text{ bytes} \tag{4}$$

Verification requires only the proof π_i, the commitment C_i, and the root C_{root}:

$$\text{Verify}(\pi_i, C_i, C_{\text{root}}) \in \{\text{true}, \text{false}\} \tag{5}$$

3.3.4 Temporal Root Chain

For historical verification, we maintain a chain of roots:

$$R_t = H(C_{\text{root}}^{(t)} \| R_{t-1} \| t) \tag{6}$$

This enables an auditor to verify the knowledge state at any historical timestamp.

3.4 Layer 3: Provenance DAG

3.4.1 Motivation

Agentic AI systems reason iteratively: they think, act (query knowledge), observe results, and reason further. Current systems produce flat text logs that provide no cryptographic guarantee that:

- Step 3 actually used the result of Step 1 (causal linkage)
- The knowledge queried in Step 2 was from the verified knowledge base (knowledge provenance)
- The final conclusion was derived from the recorded steps and not fabricated (output integrity)

3.4.2 Formal Definition

A *Provenance DAG* is a directed acyclic graph $\mathcal{P} = (N, D)$ where each node $n_i \in N$ represents a reasoning step:

$$n_i = \langle \text{id}, \text{type}, \text{content}, \text{kg_proofs}, \text{deps}, h_i \rangle \tag{7}$$

where:

- $\text{type} \in \{\text{thought}, \text{action}, \text{observation}, \text{conclusion}\}$
- content: the reasoning step's output (text)
- kg_proofs: set of Verkle proofs for knowledge accessed in this step
- $\text{deps} \subseteq N$: set of prior nodes this step depends on
- $h_i = H(\text{content} \| \text{kg_proofs} \| \{h_j\}_{j \in \text{deps}})$: cryptographic hash linking this node to its inputs

Edges $D \subseteq N \times N$ represent causal dependencies: $(n_j, n_i) \in D$ means node n_i's reasoning was influenced by node n_j's output.

3.4.3 Properties

The Provenance DAG provides three cryptographic guarantees:

1. **Tamper evidence:** Modifying any node's content invalidates its hash and all descendant hashes.
2. **Completeness:** Every knowledge access is linked to a Verkle proof; every reasoning step is linked to its causal predecessors.
3. **Non-repudiation:** The system cannot deny that a specific reasoning chain produced the final decision.

4 Unified Audit Protocol

4.1 Audit Record Structure

Every AI decision produces a complete audit record:

$$\mathcal{A} = \langle q, t, C_{\text{root}}^{(t)}, \mathcal{P}, \mathcal{V}, r, \sigma \rangle \tag{8}$$

where q is the query, t is the timestamp, $C_{\text{root}}^{(t)}$ is the Verkle root at time t, $\mathcal{P}$ is the Provenance DAG, $\mathcal{V} = \{\pi_i\}$ is the set of Verkle proofs used, r is the final response, and σ is a digital signature over the entire record.

4.2 Independent Verification

An auditor independently verifies a decision by:

1. Verify each Verkle proof π_i against $C_{\text{root}}^{(t)}$.
2. Verify the temporal chain links $C_{\text{root}}^{(t)}$ to a trusted anchor.
3. Recompute every Provenance DAG node hash h_i from its recorded content, proofs, and dependency hashes.
4. Verify the digital signature σ.
5. Confirm the DAG is acyclic and the conclusion node's content matches r.

4.3 Selective Disclosure

Using Verkle proofs, the system can prove specific knowledge was used without revealing the entire knowledge base. In healthcare, this means proving a drug interaction check was performed without exposing the patient's complete medication list.

5 Experimental Evaluation

5.1 Dataset: DrugBank 6.0

We evaluate on healthcare drug interaction detection using DrugBank 6.0 [Knox et al., 2024]:

- 4,563 FDA-approved drugs
- 1,413,413 drug-drug interactions with severity ratings
- 2,475 drug-food interactions
- Mapped to ICD-10, SNOMED-CT clinical codes

We construct a hypergraph with:

- **Pairwise edges (E_2):** Direct drug-drug interactions from DrugBank
- **Hyperedges (E_h):** Multi-drug polypharmacy risks constructed by combining interactions that share common metabolic pathways (CYP450 enzymes) or therapeutic categories

5.2 Evaluation Scenarios

5.2.1 Scenario 1: Proof Size Comparison

Table 1: Proof size comparison: Merkle vs Verkle for varying knowledge base sizes

Subgroups (k)	Merkle Proof	Verkle Proof	Reduction
64	192 bytes	96 bytes	50.0%
256	256 bytes	96 bytes	62.5%
1,024	320 bytes	96 bytes	70.0%
10,000	448 bytes	96 bytes	78.6%
100,000	544 bytes	96 bytes	82.4%
1,000,000	640 bytes	96 bytes	85.0%

Verkle proof size remains constant regardless of knowledge base size, while Merkle proofs grow logarithmically.

5.2.2 Scenario 2: Hypergraph vs Pairwise KG for Drug Safety

We evaluate whether hyperedges catch risks that pairwise edges miss, using 500 curated multi-drug prescription scenarios:

Table 2: Drug safety detection: Pairwise KG vs Hypergraph

Knowledge Representation	Precision	Recall	F1	Multi-factor Risks Caught
Pairwise KG (standard)	91.2%	78.4%	84.3%	0/47
Pairwise KG + rule combination	89.8%	85.1%	87.4%	23/47
Hypergraph (ours)	**92.5%**	**93.8%**	**93.1%**	**44/47**

The hypergraph representation catches 93.6% of multi-factor risks (44/47) that involve three or more interacting entities, compared to 0% for raw pairwise KG and 48.9% for rule-augmented pairwise KG.

5.2.3 Scenario 3: Provenance DAG Completeness

For 200 drug interaction queries processed through the agentic reasoning loop:

Table 3: Provenance DAG verification metrics

Metric	**Result**
Queries processed	200
Avg. reasoning iterations per query	3.4
Avg. DAG nodes per query	7.2
Avg. Verkle proofs per query	4.1
All DAG hashes verified	200/200 (100%)
All Verkle proofs verified	200/200 (100%)
Full audit records independently verifiable	200/200 (100%)

5.2.4 Scenario 4: Computational Overhead

Table 4: VHP computational overhead per query

Operation	**Time (ms)**	**Overhead**
Hypergraph query (subgraph extraction)	1.2	Baseline
Verkle proof generation	0.8	+1.1%
Verkle proof verification	0.3	+0.4%
Provenance DAG node creation	0.2	+0.3%
Provenance DAG hash computation	0.5	+0.7%
Audit record generation	1.5	+2.0%
Total VHP overhead per query	**3.3**	**+3.4%**

5.2.5 Scenario 5: Tamper Detection

We simulate tampering by modifying 1, 10, and 100 hyperedges in the knowledge base:

Table 5: Tamper detection performance

Tampered Hyperedges	**Detection Rate**	**Detection Time**	**False Positives**
1	100%	0.4 ms	0
10	100%	0.5 ms	0
100	100%	0.8 ms	0

6 Discussion

6.1 Why Three Layers Are Necessary

Each layer addresses a distinct gap that the others cannot fill:

- **Remove Hypergraph:** Multi-factor polypharmacy risks (44/47 in our evaluation) go undetected. Pairwise KGs check each drug pair independently and miss combinatorial effects.
- **Remove Verkle:** Proof sizes grow with knowledge base scale. At 1M subgroups, Merkle proofs are 640 bytes vs. Verkle's constant 96 bytes. For audit trails with 20+ proofs per decision, this is the difference between feasible and infeasible real-time audit.
- **Remove Provenance DAG:** You can prove *what* knowledge was used but not *how* the agent reasoned. An auditor cannot trace causal influence between reasoning steps.

6.2 Reasoning-Engine Agnosticism

VHP does not inspect or constrain the reasoning engine. It wraps inputs (verified knowledge context) and outputs (reasoning step content) with cryptographic metadata. This means VHP can be applied to:

- LLMs (GPT-4o, Claude) for high-accuracy reasoning
- SLMs (Phi-3, Qwen-2.5) for cost-efficient specialized tasks
- Multi-agent systems (AutoGen, CrewAI) where each agent's reasoning is tracked
- Any future reasoning architecture

6.3 Practical Deployment

For production deployment, the Verkle root can be anchored to a blockchain (Ethereum, Hyperledger) for immutable timestamping, providing additional trust guarantees without requiring the full knowledge base to be on-chain.

6.4 Limitations

1. **Hypergraph construction:** Multi-way interactions must be curated or mined from clinical data. Automated hyperedge discovery from drug labels remains future work.
2. **Verkle implementation maturity:** Verkle tree libraries are less mature than Merkle tree implementations. Our prototype uses a simplified polynomial commitment scheme; production systems should use Pedersen commitments over Bandersnatch curves.
3. **Prototype reasoning engine:** We use simulated domain reasoning rather than actual LLM inference. The verification architecture is engine-agnostic by design; swapping in a real LLM requires no changes to Layers 1–3.
4. **Privacy:** While Verkle proofs enable selective disclosure, full privacy-preserving audit trails would require integration with zero-knowledge proofs, which we leave as future work.

6.5 Future Work

Key extensions include: (1) zero-knowledge proof integration for privacy-preserving audit trails, (2) blockchain anchoring for immutable root timestamps, (3) automated hyperedge discovery from clinical trial data and FDA drug labels, and (4) real-world deployment study with clinical decision support systems.

7 Conclusion

We introduced VHP, a three-layer verification architecture that provides end-to-end cryptographic guarantees for AI decision systems. By combining hypergraph knowledge representation (capturing multi-way domain interactions), Verkle tree verification (constant-size integrity proofs), and Provenance DAG tracking (causal reasoning chains), VHP enables independently

verifiable AI decisions with less than 4% computational overhead. Our evaluation on healthcare drug interaction detection demonstrates that VHP catches 93.6% of multi-factor risks that pairwise knowledge graphs miss, while maintaining complete audit trail verifiability.

As AI systems increasingly operate in regulated, safety-critical domains, the ability to cryptographically verify *what the system knew*, *what combinations it checked*, and *how it reasoned* is essential. VHP provides a principled, model-agnostic solution to this challenge.

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